

# Convergence Analysis

## 1 Analysis

We bound the loss with the following inequality, derived in the appendix.

$$L(w) = S(w) + a * R(x_i) + b \quad (1)$$

Theorem 1. The estimator is consistent under the stated assumptions and converges to the true parameter as the sample size grows without bound.

The remaining discussion is ordinary body prose following the theorem statement and should be extracted as a paragraph in reading order.